

JEE MAIN REVISION MASTER: SIMPLE HARMONIC MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kinematics & Dynamics of SHM	8 PYQs	SHM Basics
Moderate	Energy in SHM & Spring-Pendulum Systems	3 PYQs	Energy & Systems

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Kinematics & Dynamics of SHM	$x(t) = A\sin(\omega t + \phi)$, $v = \omega\sqrt{A^2 - x^2}$, $a = -\omega^2 x$, $T = 2\pi\sqrt{\frac{m}{k}}$
Energy in SHM & Spring-Pendulum Systems	$E_{\text{total}} = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2 = \text{const}$, $T_{\text{spring}} = 2\pi\sqrt{\frac{m}{k}}$, $T_{\text{pendulum}} = 2\pi\sqrt{\frac{l}{g}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kinematics & Dynamics of SHM (High Yield — 8 PYQs)

Core Principle: Differential equation of SHM ($d^2x/dt^2 + \omega^2 x = 0$), displacement, velocity, acceleration, and phase relations.

■ **Caution / JEE Pitfall:** Velocity leads displacement by $\pi/2$; acceleration leads displacement by π rad (opposite phase).

■ **Speed Trick:** At mean position ($x = 0$): $v_{\text{max}} = A\omega$, $a = 0$. At extreme ($x = \pm A$): $v = 0$, $a_{\text{max}} = \omega^2 A$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A mass of 1 kg is executing SHM. Its displacement is given by $x=6.0\cos(100t+\pi/4)$ cm. What is the maximum kinetic energy?..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The variations of kinetic energy $K(x)$, potential energy $U(x)$ and total energy as a function of displacement of a particle in SHM is as ..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "For a particle executing simple harmonic motion (SHM), at its mean position..." [Ans: N/A]

3.2 Energy in SHM & Spring-Pendulum Systems (Moderate Yield — 3 PYQs)

Core Principle: Energy oscillation in SHM (frequency of K and U is $2f$), springs in series/parallel, and simple pendulum period.

■ **Caution / JEE Pitfall:** Kinetic energy and potential energy oscillate at TWICE the frequency (2ω) of displacement.

■ **Speed Trick:** When a spring of constant k is cut into ratio $1 : n$, the parts have constants $k_1 = k(n+1)$ and $k_2 = k(n+1 / n)$.

Representative Source PYQs:

- **[2021 Shift] Q#2 (MCQ):** "A pendulum oscillates simple harmonically and only if l. the sizer of the bob of pendulum is negligible in comparison with the length of the..." **[Ans: N/A]**
- **[2020 Shift] Q#3 (MCQ):** "A tray of mass 12kg is supported by two identical springs as shown in figure. When the tray is pressed down slightly and then released, it..." **[Ans: N/A]**
- **[2017 Shift] Q#1 (MCQ):** "Two simple pendulums A and B are made to oscillate simultaneously and it is found that A completes 10 oscillations in 20 sec and B c..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A mass of 1 kg is executing SHM. Its displacement is given by $x = 6.0 \cos(100t + \pi/4)$ cm. What is the maximum kinetic energy?

- (A) 3 J
- (B) 6 J
- (C) 9 J
- (D) 18 J

Correct Answer: None

Solution Analysis: For SHM, $x = A \cos(\omega t + \phi)$ Given displacement is $x = 6.0 \cos(100t + \pi/4)$ cm So the amplitude is $A = 6.0 \text{ cm} = 0.06 \text{ m}$ and angular frequency is $\omega = 100 \text{ rad s}^{-1}$ In SHM, maximum kinetic energy is equal to the total mechanical energy: $K_{(\max)} = (1/2)m\omega^2 A^2$ Substitute the values: $K_{(\max)} = (1/2)(1)(100)^2(0.06)^2$ $K_{(\max)} = (1/2) \times 10000 \times 0.0036$ $K_{(\max)} = 18 \text{ J}$ So the maximum kinetic energy is 18J Correct option: Option D.

Benchmark Problem 2: KCET 2017 (Q#1)

Two simple pendulums A and B are made to oscillate simultaneously and it is found that A completes 10 oscillations in 20 sec and B completes 8 oscillations in 10 sec . The ratio of the length of A and B is

- (A) (25 / 64)
- (B) (64 / 25)
- (C) (8 / 5)
- (D) (5 / 4)

Correct Answer: None

Solution Analysis: To determine the ratio of the lengths of pendulums A and B , we start with the given information: For pendulum A : It completes 10 oscillations in 20 seconds. The time period $T_{(A)}$ is the total time divided by the number of oscillations: $T_{(A)} = (20 \text{ seconds} / 10) = 2 \text{ seconds}$ The formula for the period of a pendulum is given by: $T_{(A)} = 2\pi\sqrt{\frac{L_{(A)}}{g}}$ where $L_{(A)}$ is the length of pendulum A and g is the acceleration due to gravity. For pendulum B : It completes 8 oscillations in 10 seconds. The time period $T_{(B)}$ is: $T_{(B)} = \frac{10 \text{ seconds}}{8} = 1.25 \text{ seconds}$